

Can Prerequisite Knowledge Be Recovered from Research Paper Abstracts?

A Graph-Based Proof-of-Concept

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Abstract - Research abstracts often assume background knowledge that they do not explain. I study whether this missing prerequisite knowledge can be recovered from an abstract alone. My proof-of-concept combines a SciBERT concept extractor fine-tuned on STEM-ECR, an off-the-shelf sentence-embedding linker to the LectureBank prerequisite graph, and backward graph traversal. Only SciBERT is trained. On a hand-built gold set of 12 ML/NLP papers with 100 annotated prerequisites, I first measure the premise itself: 39% of prerequisites are stated verbatim and 76% under the primary semantic rule. The extractor reaches entity-level F1 0.765 in-distribution, compared with 0.291 for a noun-phrase baseline. However, graph traversal does not improve end-to-end performance. It triples the number of predictions for only +0.008 recall, while precision falls from 0.491 to 0.148 and F1 falls to 0.209, below the no-graph baseline of 0.539. Failure analysis shows that 43.8% of missed prerequisites are likely linking failures, while only 7.8% are traversal failures. The main finding is therefore that short-phrase entity linking is a larger bottleneck than graph traversal.

Index Terms - concept extraction, prerequisite knowledge graphs, entity linking, scientific NLP, proof of concept, negative results.

I. INTRODUCTION

A. Motivation

Reading an unfamiliar research paper is difficult partly because the paper assumes that the reader already knows some of the background. For example, an abstract mentioning a “reparameterisation of the variational lower bound” assumes knowledge of variational inference, latent-variable models, and stochastic gradient estimation. A system that could identify this assumed background from the abstract could help with tasks such as ordering course topics, building reading lists, and helping students enter an unfamiliar research area.

B. Problem of Study

I study the following task: given only a research paper abstract, identify the background concepts a reader should already understand to follow it. I define the target reader as a first-year ML/NLP graduate student so that the meaning of “prerequisite” is fixed. A prerequisite is a concept, method, or mathematical tool that the paper builds on, rather than the paper’s own contribution or newly introduced terminology. I use only abstracts because they are freely available and generally short enough to fit within BERT’s 512-token input limit [1]. The work is limited to ML/NLP papers and is presented as a proof-of-concept rather than a finished tool.

C. Knowledge Gap and Contributions

Existing resources provide only parts of what this task requires. STEM-ECR [2] provides annotations for scientific concepts that are present in text, so a model trained on it is fundamentally extractive. LectureBank [3] provides prerequisite relationships between concepts, but assumes that the concepts are already known. Neither resource directly maps an abstract to prerequisite knowledge that may be absent from the abstract.

I therefore combine the two resources instead of creating a new large prerequisite dataset. The system first extracts concepts stated in the abstract, links them to LectureBank nodes, and then traverses the graph backward to find

upstream concepts. My main contribution is the experimental evaluation of whether this composition actually helps. I quantify how much prerequisite information is present in abstracts, evaluate the SciBERT extractor, test the cleanup and graph stages, and analyze the causes of end-to-end failures.

D. Summary of Methodology, Experiments and Results

SciBERT [5] is fine-tuned for BIO concept extraction on 110 STEM-ECR abstracts. The output is cleaned using rule-based filters, linked to 208 LectureBank concepts using all-MiniLM-L6-v2 [4], and expanded through 921 positive prerequisite edges. On 12 hand-annotated papers, the graph-based variants perform worse than simply using the cleaned extracted concepts: the best traversal variant obtains F1 0.209 versus 0.539 for the no-graph baseline. The failure analysis shows that linking is the main bottleneck. The result is negative for the central hypothesis, but the experiments identify where the pipeline fails and provide a clear direction for improvement.

II. METHODOLOGY

The pipeline has four stages: EXTRACT concepts stated in the abstract, CLEAN the extractor output, LINK each phrase to a LectureBank node, and TRAVERSE the graph backward to obtain upstream prerequisites. SciBERT is the only trained component; the linker and graph are used off the shelf.

A. Stage 1 - Concept Extraction

STEM-ECR [2] contains 110 abstracts across 10 scientific domains (11 each), with 6,167 annotated entity spans in brat standoff format. I verified the character offsets before writing any conversion code: `txt[start:end]` reproduced the recorded surface form for all 6,163 continuous spans, so labels could be assigned by offset overlap rather than by error-prone string matching. The seven annotation types (Process, Material, Data, Method, and residual Result/Object/Task) are collapsed into a single CONCEPT class because the task only requires identifying where concepts are, not what type they are. Labels are assigned

document-wide and then split into sentences, so the 3 spans that cross a line break survive; the 4 discontinuous spans are labelled fragment-wise; and overlapping spans are resolved longest-first (146 dropped). The counts reconcile exactly: 6,021 spans labelled plus 3 line-break splits give the 6,024 entities recovered.

BIO labels are projected onto SciBERT word pieces using the standard convention: -100 on special tokens ([CLS], [SEP]) and on sub-word continuations - so each word contributes exactly one prediction, and the loss ignores the rest. Because this projection is the step most likely to corrupt training data silently, I inspected one worked example token-by-token to confirm the alignment before training. I fine-tune `allenai/scibert_scivocab_uncased` for 10 epochs with learning rate $3e-5$ and batch size 16 using HuggingFace Transformers [6], selecting the best epoch by validation F1. The train/validation split is document-level and stratified by domain (90/20 documents); a sentence-level split would place sentences from the same abstract on both sides and inflate the reported score, so I avoided it deliberately.

B. Stage 2 - Cleanup

At inference, the extractor is applied to ML/NLP abstracts outside its STEM-ECR training distribution and produces 398 raw spans across the 12 gold papers, including verbs, metrics, and generic words. Five rule-based operations are applied uniformly: repair tokenizer spacing, split coordination, strip leading determiners, remove number- or metric-dominated spans and bare non-concept tokens, and deduplicate. Every removal is logged. The rules do not remove a phrase simply because it is the paper’s own contribution; deciding whether something is a prerequisite is left to the downstream graph.

C. Stage 3 - Linking

Each cleaned phrase is embedded with `all-MiniLM-L6-v2` [4] and matched to the nearest of 208 LectureBank concepts. A match is accepted only when cosine similarity passes a frozen threshold. The threshold was selected by manually auditing matches before downstream scores were considered. Each graph node is indexed by its name, its Wikipedia page title where available (184/208 nodes), and a possessive-stripped variant; the maximum similarity over these forms is used.

D. Stage 4 - Traversal

LectureBank contains 42,750 annotated concept pairs; I retain the 921 positive prerequisite edges and build a directed graph with NetworkX [7]. Edge direction was verified empirically rather than assumed from column names: `linear algebra` has out-degree 98 and in-degree 0, while concepts such as `Neural Machine Translation` have high in-degrees. Prerequisites are therefore predecessors of a concept. I test direct predecessors and the full transitive ancestor set at thresholds 0.65 and 0.60.

E. Evaluation Protocol

The gold set contains 12 well-known ML/NLP papers [1], [8]-[16], [18], [19]] and 100 hand-written prerequisites. I

chose landmark papers deliberately, because their prerequisites are widely agreed upon and so the hand annotation is defensible rather than idiosyncratic. The gold set is used only for evaluation, never for training. Prerequisites are grouped into A/B/C bands by how well I expected the graph to cover them, so that any breakdown between in-coverage and out-of-coverage papers would be visible. Because gold wording and graph names can differ, I use three matching rules: strict verbatim matching after normalization, semantic matching when both phrases map to the same node or have cosine similarity at least 0.60, and lenient matching using at least 60% content-word overlap. Semantic matching is the primary rule.

Scoring is at the set level and macro-averaged over papers: recall is the fraction of gold phrases matched by at least one prediction, and precision the fraction of predictions matching at least one gold phrase. I use this set-level bipartite match rather than a strict one-to-one assignment because gold phrases are often compound (“Convolutional neural networks (CNNs)”) while graph nodes are atomic, and a one-to-one assignment would penalise the system for the granularity of the gold set rather than for genuine errors.

III. EXPERIMENTS AND RESULTS

A. Extraction Quality (in-distribution)

TABLE I. ENTITY-LEVEL SCORES, STEM-ECR VALIDATION (20 HELD-OUT DOCUMENTS)

System	P	R	F1
POS chunker (baseline)	0.268	0.319	0.291
SciBERT fine-tuned	0.740	0.791	0.765

SciBERT achieves precision 0.740, recall 0.791, and F1 0.765 on the 20 held-out STEM-ECR documents, compared with 0.268, 0.319, and 0.291 for the POS noun-phrase baseline. This is a +0.473 F1 improvement. Validation loss rises after epoch 3 while F1 plateaus, indicating mild overfitting on 1,166 training sentences; selecting by validation F1 handles this. Computer Science is the best of the ten domains at F1 0.832, although this is based on only 2 validation documents and 93 entities.

B. How Much Background Is Actually Absent?

TABLE II. FRACTION OF THE 100 GOLD PREREQUISITES STATED IN THE ABSTRACT

Match rule	Extractable ceiling
strict (verbatim)	0.39
semantic (primary)	0.76
lenient (word overlap)	0.71

This experiment tests the main assumption behind the project. Depending on the matching rule, between 24% and 61% of the true prerequisites do not appear in the abstract in a recognisable form. This supports testing a graph-based approach because extraction alone cannot recover those missing concepts. The semantic score being higher than the lenient score was unexpected. Manual inspection of the 16 semantic-only matches found genuine paraphrases such as

“LSTM networks” versus “Long Short Term Memory” and “phrase-based statistical machine translation” versus “phrase-based SMT system.” Therefore, lenient word overlap is not an upper bound, and all three rules are reported.

C. Cleanup and Signal Preservation

Cleanup reduces 398 spans to 305, a 23% reduction (42 to 30 for ResNet). To check whether useful concepts are removed, mention recall is measured before and after cleanup against a fixed noun-phrase ceiling. The removal rules cut 25% of spans while recall remains 0.934, and the rewriting rules raise recall to 1.000. No gold prerequisites are lost. This supports describing the cleanup as audited. The 1.000 recall should only be interpreted as evidence that real concepts were not deleted, not as an overall quality score.

D. Linking, Traversal, and End-to-End Result

At the frozen threshold, 74 of 305 cleaned phrases (24.3%) resolve to a graph node, giving 56 entry points at $\tau=0.65$ and 73 at $\tau=0.60$. The graph itself has 210 nodes - the 208 named concepts plus two ids that appear in the edge file but have no entry in the topic list - along with 921 edges and 9 isolated nodes. Contrary to my initial assumption, it is not a DAG: it contains 24 simple cycles (12 of length 2, 8 of length 3, 4 of length 4), for example dimensionality reduction $\leftrightarrow$ SVD, and a 4-cycle over ROUGE / BLEU / Neural Machine Translation / IBM Translation Models, reflecting annotation disagreements about edge direction. Transitive closure is still well-defined, so no change to the traversal algorithm was needed. Traversal produces between 14.8 and 21.6 upstream concepts per paper, against a gold average of 8.3, so it clearly adds volume; the question is whether that volume is useful.

There is some evidence that it is not merely returning a generic foundation shared by every paper: only 1 concept is common to all 12 papers under direct traversal, and 3 under full ancestors. And for individual papers the output can look right - for the Transformer paper, full-ancestor traversal returns Seq2seq, Cross entropy, Backpropagation, Gradient descent, Word Embedding, and Word2Vec - all recognisable background, none of it stated in that abstract. This is the mechanism working as intended in isolation, which makes the aggregate result below more informative.

TABLE III. FULL PIPELINE VS GOLD PREREQUISITES (SEMANTIC RULE, MACRO-AVERAGED)

System	Preds	P	R	F1
cleaned concepts (no graph)	25.4	0.396	0.841	0.539
entry points only, $\tau=0.60$	6.1	0.491	0.348	0.408
entry + direct, $\tau=0.60$ (best)	22.2	0.148	0.356	0.209
entry + ancestors, $\tau=0.60$	27.7	0.124	0.356	0.184

Adding traversal does not improve the result. It roughly triples the number of predictions from 6.1 to 22.2 per paper for only +0.008 recall, while precision falls from 0.491 to 0.148. Every traversal variant is below the no-traversal baseline, and the cleaned extraction baseline has the best F1 at 0.539. This ranking is stable across all three matching

rules. Direct predecessors outperform full ancestors everywhere, the two thresholds differ by only 0.001 F1 end-to-end, and Band C is not measurably worse than Band A (0.218 vs. 0.184).

IV. EVIDENCE FOR CLAIMED CONTRIBUTIONS

A. Claim 1 - The premise is real and quantified

Table II shows that under the primary semantic rule, 24 of 100 gold prerequisites are not stated in the abstract, rising to 61 under strict verbatim matching. Section III-D also shows that graph traversal can supply some absent concepts, as demonstrated by the Transformer example.

B. Claim 2 - The extractor is sound and cleanup is safe

Table I establishes extraction quality against a real, non-trivial baseline on a held-out, document-level, domain-stratified split. The cleanup ablation (Section III-C) shows that the removal rules cut 25% of spans without reducing recall, providing measured rather than asserted evidence that they remove noise without deleting real concepts. I am also careful about what this evidence does and does not show: the mention-recall metric saturates at 1.000, so it supports exactly one proposition - that the filter did not delete real concepts and must not be read as an overall quality score for the extractor.

C. Claim 3 - Linking is a larger bottleneck than traversal

TABLE IV. FAILURE BREAKDOWN, BEST VARIANT

Reason	Count	Share
gold-side link miss (node likely exists)	28	43.8%
absent from graph	26	40.6%
linking miss (wrong/no node)	5	7.8%
graph miss (traversal never reached)	5	7.8%

Of 64 missed gold prerequisites, 28 (43.8%) are gold-side link misses where the graph node likely exists, 26 (40.6%) are absent from the graph, 5 (7.8%) are linking misses involving a wrong or missing node, and 5 (7.8%) are graph misses where traversal never reaches the concept. Only 7.8% are directly attributable to traversal. An earlier diagnostic counted 84% as graph-absent because it only checked whether the gold phrase had a node above threshold. Checking the actual best-node scores showed that many of these were linker failures, reducing the apparent coverage problem by roughly half.

A concrete case illustrates this issue. LectureBank stores the convolutional-network node as CNN's. The phrase “convolutional neural networks” scores only 0.469 against it, below threshold, but 0.673 against “graph convolutional networks,” causing the wrong node to be selected. This affects 5 gold prerequisites across 4 papers. Indexing graph nodes using Wikipedia page titles increases the correct similarity to 0.926. Since top-1 nearest-neighbour linking always returns a result, such errors can be silent rather than producing an explicit failure.

V. DISCUSSION / LIMITATIONS

A. Why the composition underperformed

The most important reason is a subtle interaction between the gold set and the method. The gold prerequisite lists were written while reading the abstracts, so their wording is naturally anchored to the abstracts' wording, and the 0.76 semantic ceiling therefore partly measures that anchoring rather than a property of abstracts in general. A gold set that mostly names concepts the abstract already states will, almost by construction, reward extraction and leave the graph little room to add value - which is close to what the end-to-end numbers show. What the graph does add is a foundational tail (linear algebra, probability, matrix multiplication) that is arguably genuine background for any ML paper but appears in almost none of the hand-written lists, so it is scored as noise even where it is defensible. The founding premise is therefore only half-supported: a real fraction of prerequisites is genuinely absent from abstracts, but backward traversal through LectureBank does not reliably recover those particular concepts, and the concepts it does add fall outside what the gold set rewards.

B. Scale and Annotation

The gold set contains 12 papers and 100 prerequisites, annotated by one author using the written definition. There is no second annotator or inter-annotator agreement, and training used one random seed. As a proof-of-concept, the absolute scores should be interpreted as showing the direction of the results rather than precise estimates at larger scale.

C. Resource Mismatch

STEM-ECR is a general-STEM dataset with only 11 Computer Science abstracts, and its Process/Material/Data scheme is not identical to an ML concept annotation. The extractor is therefore used out of distribution on the target ML/NLP abstracts. LectureBank contains 208 concepts and is designed as an NLP teaching graph, so coverage is limited for areas such as vision, theory, and contrastive learning. No token-level labels exist for the target ML abstracts, so out-of-distribution extraction precision cannot be measured directly; only recall-style proxies are available.

D. Future Work

The failure breakdown identifies short-phrase entity linking as the clearest target for improvement, since 44% of misses are associated with this issue. Possible improvements include acronym expansion, alias tables, context-aware phrase embeddings, and reranking the top-k candidate nodes instead of selecting only the top-1 neighbour. A second direction is to build a gold set without first reading the abstract, reducing anchoring bias and providing a fairer test of whether graph traversal can recover prerequisite knowledge that is actually absent from the abstract.

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USE OF AI TOOLS

In accordance with the course's permitted-use policy, the research design, the problem framing, the three-stage architecture, the choice of datasets and resources (STEM-ECR, LectureBank), the evaluation protocol, the banded gold set, and the interpretation of all results - was my own. AI assistance was used in three bounded ways: as a coding aid for some of the implementation in Google Colab (for example, the BIO-conversion and tokenizer-alignment scripts and training boilerplate), which I reviewed, ran, and debugged; as a sounding board for methodological choices such as freezing thresholds before evaluation and designing the cleanup ablation, where the final decisions and their justification were mine; and for language editing of this report. All experimental results and numerical values were produced and verified by me from my own runs.